GROUP	NORM	LL TOP	LL BOTTOM
BIGGEST BY NORM	1.212	Paris, Paris, Marse	ufact, Logged, otomy
	0.352	Paris, Paris, Marse	ufact, Spanish, Gerr
	0.297	Paris, Paris, Copenhagen	ceremonial, cade, uana
MEDIUM BY NORM	0.033	ufact, Gerr, sheriff	Paris, Paris, ienne
	0.033	Logged, turtle, ceremon	Paris, Paris, France
	0.033	ufact, sec, recess	Paris, Paris, qus
SMALLEST BY NORM	0.001	,, the, and	VIDIA, advertisement, Dialogue
	0.001	,, the, and	VIDIA, advertisement, Magikarp
	0.001	,, the, and	VIDIA, advertisement, Companies
ZERO VECTOR	0	,, the, and	VIDIA, advertisement, Companies

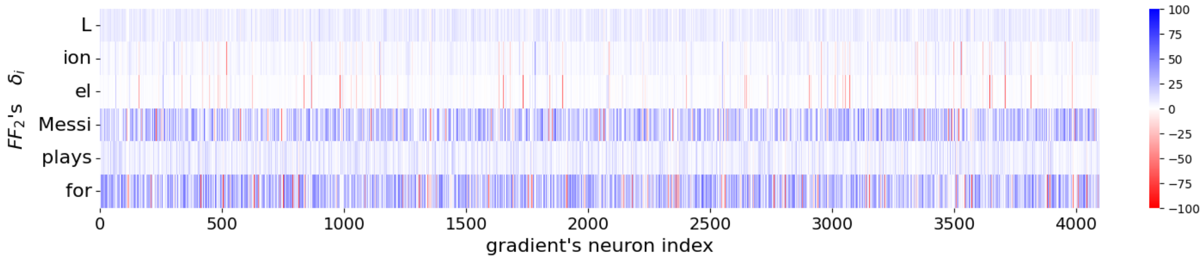
Table 2: Sample of gradient's neurons projection via Logit Lens (LL) from GPT2-medium, FF_2 matrix, layer 14. **LL TOP** stands for the most probable tokens via LL, while **BOTTOM** are the most improbable ones. In this example, we edit the prompt "Lionel Messi plays for" with the editing target "Paris". In the projected tokens we notice the predominance of "Paris", and also that gradient's neurons with a relatively low norm project the same tokens as the zero-vector.

PROMPT TOKEN	NORM	LL TOP	LL BOTTOM
L	0.029	Rutherford, Apost, PROG	Paris, Paris, ienne
ion	0.035	ramid, ngth, livest	France, ée, É
el	0.067	unlaw, owship, arantine	Libyan, Libya, France
Messi	0.165	ğ, relic, ejected	vu, igmat, tain
plays	0.026	surv, POV, NTS	Paris, hotels, Merit
for	0.165	ceremonial, ado,	Paris, Paris, Copenhagen

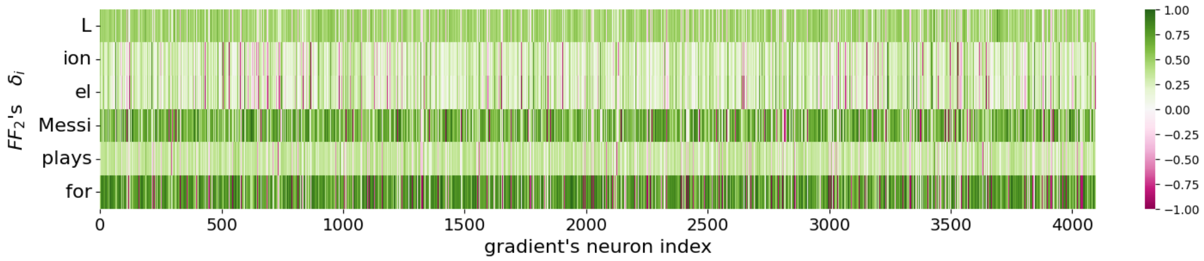
Table 3: The Logit Lens of the VJPs (δ_i) of GPT2-medium, FF_2 matrix, layer 14. Notice the dominance of "Paris" (the editing target) in the projected vocabulary and the norm ratio of the vectors.

GROUP	NORM	LL TOP	LL BOTTOM
BIGGEST BY NORM	0.438	Cruz, ization, ize	mathemat, trave, nodd
	0.422	Football, Jr, Team	theless, challeng, neighb
	0.369	psychiat, incent, theless	Jr, Junior, Sr
MEDIUM BY NORM	0.02	the, a, hire	irez, inelli, intosh
	0.02	the, a, one	theless, Magikarp, irez
	0.02	perpend, coerc, incent	Junior, Jr, Football
SMALLEST BY NORM	0.002	,, the, and	Magikarp, VIDIA, advertisement
	0.002	,, the, and	advertisement, VIDIA, Magikarp
	0.001	,, the, and	VIDIA, advertisement, Companies
ZERO VECTOR	0	,, the, and	VIDIA, advertisement, Companies

Table 4: Samples of gradient's neurons projection via Logit Lens (LL) from GPT2-medium, FF_1 matrix, layer 14.

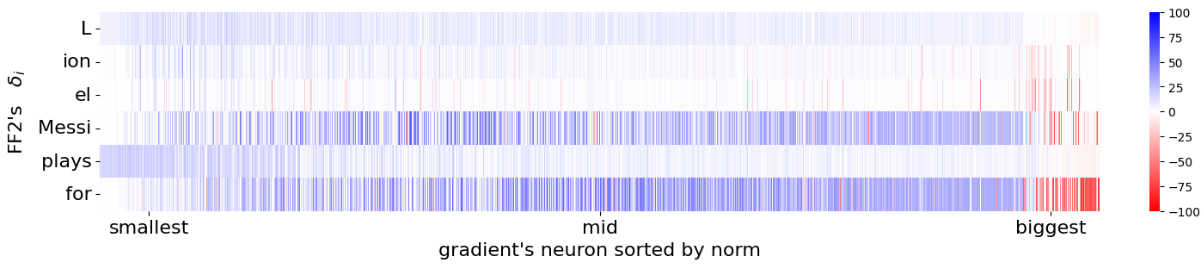


(a) Logit Lens intersection

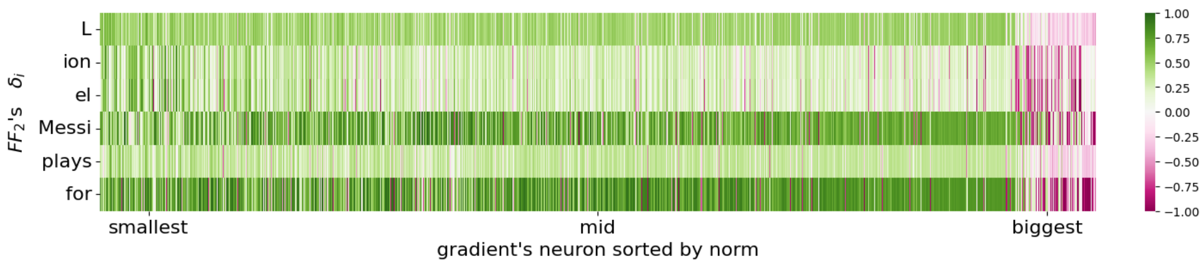


(b) Cosine Similarity

Figure 7: VJPs (δ_i, the spanning set of FF_2) and single gradient's neurons comparison for layer 14's FF_2. In Figure 7a a positive score is the amount of shared tokens between the top 100 most probable tokens after projecting each vector. A negative score is presented if multiplying one of the vectors by -1 produces a higher amount of shared tokens (than the score presented with a negative sign). The reason we apply this -1 multiplication is to address cases where two vectors have high correlation up to their sign (similarly to two vectors that overlap each other after one of them is multiplied by -1, making their cosine similarity negative).



(a) Logit Lens intersection

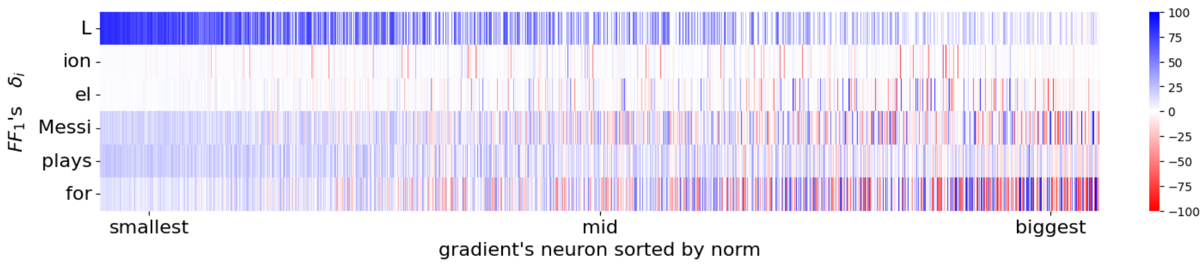


(b) Cosine Similarity

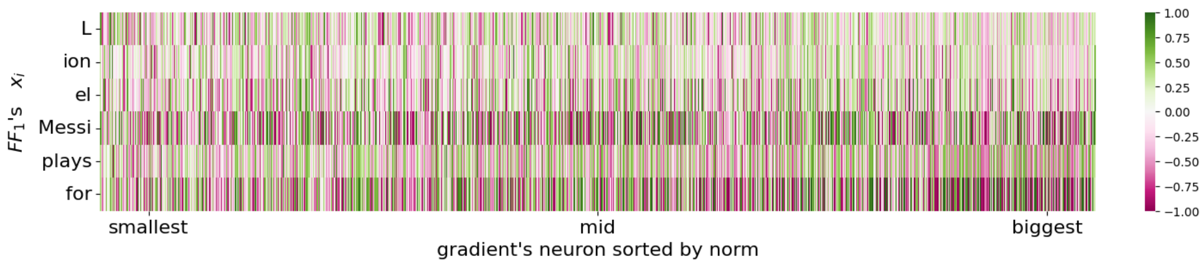
Figure 8: VJPs (δ_i) and single gradient's neurons (sorted by norm) comparison for layer 14's FF_2. It is noteworthy that high-norm neurons, corresponding to those activated with positive activations during the forward pass, inject the VJP of "for" with a flipped sign. Medium-sized norm neurons also exhibit correlation with the representative vector of "for". Smallest by-norm neurons show minimal correlation; we refer to their proximity to the zero vector.

PROMPT TOKEN	NORM	LL TOP	LL BOTTOM
L	9.499	, , , the	FontSize, 7601, Magikarp
ion	7.808	fall, fish, wood	nodd, incorpor, accompan
el	7.728	Cruz, McC, Esp	perpend, mathemat, shenan
Messi	6.944	Jr, Junior, Sr	theless, psychiat, incent
plays	6.715	football, golf, alongside	hedon, ilts, uries
for	6.596	Team, team, a	irez, newsp, Magikarp

Table 5: The Logit Lens of the inputs (x_i) of GPT2-medium, FF_1 matrix, layer 14. According to our observation Section 5.2, those are also the embeddings the gradients inject into the model's weights.



(a) Logit Lens intersection



(b) Cosine Similarity

Figure 9: The inputs x_i (which are FF_1's spanning set) and single gradient's neurons (sorted by norm) comparison for layer 14's FF_1.